Formation 10 : Tensorflow

KRYPTOSPHERE®

Environnement virtuel

```
alexandre@PCAlexandre  ~  python3 -m venv .venv
source .venv/bin/activate
```

```
alexandre@PCAlexandre  ~  pip install nom_du_package
```

Dimension du dataset

```python
print("x_train :", x_train.shape)
print("y_train :", y_train.shape)
print("x_test  :", x_test.shape)
print("y_test  :", y_test.shape)
```

```python
x_train = x_train.reshape()
x_test = x_test.reshape()
```

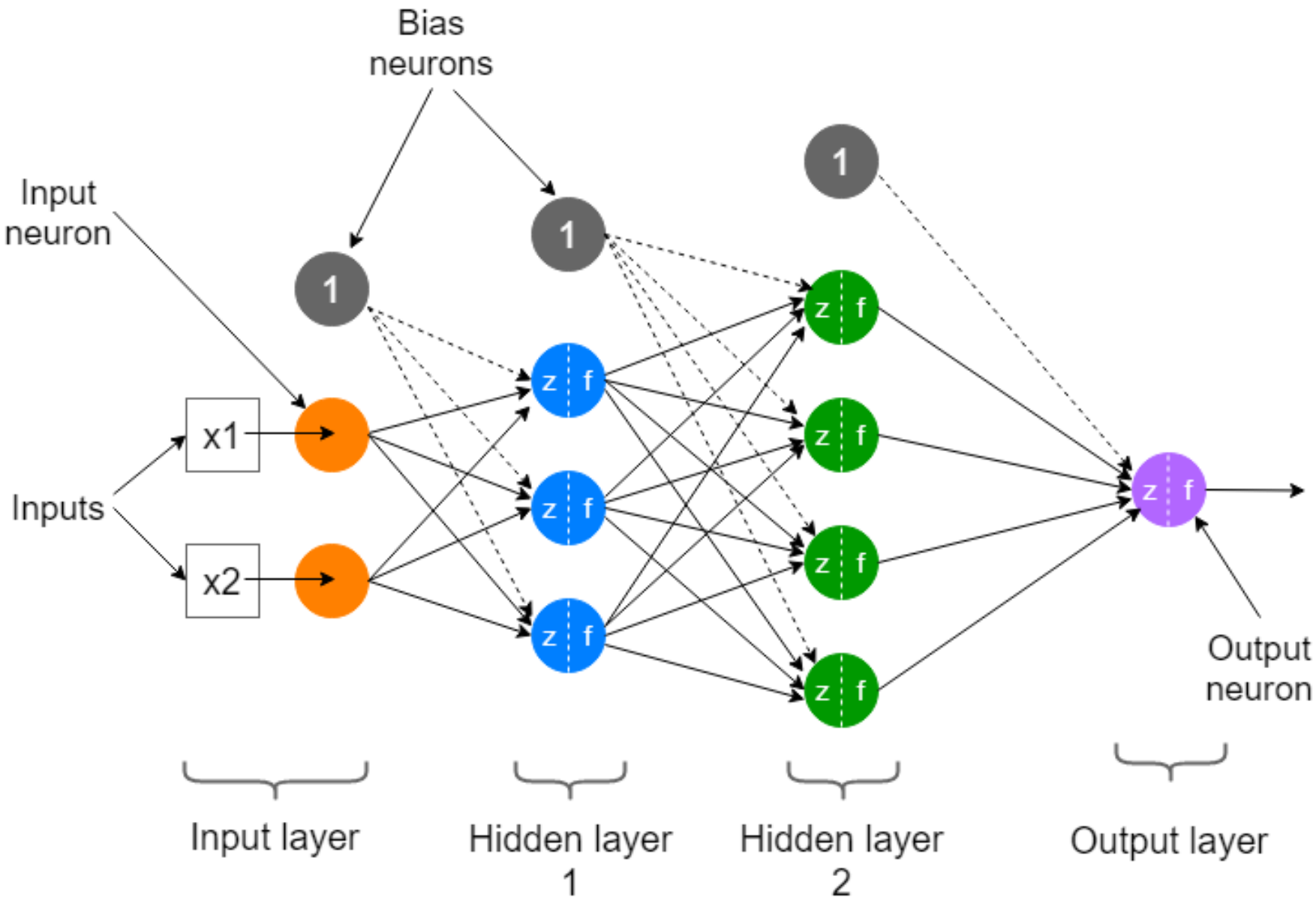
Créer le modèle

```
model = tf.keras.Sequential([
    ...
])

model.summary()
```

```
tf.keras.layers.Input(shape=(...)),
tf.keras.layers.Dense(..., activation="..."),
tf.keras.layers.Dropout(...),
tf.keras.layers.Dense(..., activation="...")
```

- Réseau de Neurone



Entrainer le modèle

```
history = model.fit(
    x_train,
    y_train,
    validation_split=...,
    epochs=...,
    batch_size=...,
    verbose=1
)
```

```
model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

Tracer les courbes d’entrainement

- plt.figure(figsize=(8, 4))
- plt.plot(history.history["loss"], label="train_loss")
- plt.plot(history.history["val_loss"], label="val_loss")
- plt.title("Loss pendant l'entraînement")
- plt.xlabel("Epoch")
- plt.ylabel("Loss")
- plt.legend()
- plt.show()
-

- plt.figure(figsize=(8, 4))
- plt.plot(history.history["accuracy"], label="train_accuracy")
- plt.plot(history.history["val_accuracy"], label="val_accuracy")
- plt.title("Accuracy pendant l'entraînement")
- plt.xlabel("Epoch")
- plt.ylabel("Accuracy")
- plt.legend()
- plt.show()

Evaluer le modèle

```
test_loss, test_accuracy = model.evaluate(x_test, y_test, verbose=0)

print(f"Test loss     : {test_loss:.4f}")
print(f"Test accuracy : {test_accuracy:.4f}")
```

A vous de jouer !

TP : entrainer un réseau de neurone sur Mnist

```
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
```